

AYBUStore: Intelligent Campus E-Commerce Ecosystem

CENG306 Term Project Final Proposal

Muhammed Yıldız (myz21), Utkan Dindaroğlu, Ahmet Selim Yılmaz,
Seyfullah Gülyazı, Süleyman Fatih Sert

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1 Abstract

AYBUStore is a specialized, AI-enhanced e-commerce platform designed to bridge the physical gap between Ankara Yıldırım Beyazıt University's distributed campuses (Esenboğa, Etlik, Bilkent, Çubuk). This report presents a comprehensive software engineering plan, utilizing a modern tech stack (Next.js 14, Firebase, LLM-based assistants) to solve the "Four-Campus Distribution Problem." We define the technical architecture, project management lifecycle using COCOMO estimation, and rigorous requirements specification. Our methodology focuses on a Hyper-Local Closed-Loop Digital Ecosystem (HL-CLDE) targeting a 30% increase in merchandise accessibility and a 1.2s Time-to-Interactive (TTI) benchmark.

2 Problem Definition & Motivation

The AYBU community lacks a centralized, real-time digital channel for branded apparel and faculty-specific resources. Students at distant campuses like Çubuk or Esenboğa face high search and transportation costs to access physical stores in Etlik. Existing generic marketplaces lack campus-specific logistics, department-targeted item filtering, and AI-driven personalization tailored to student life. This project is motivated by the need for institutional digital identity and operational efficiency in campus commerce.

3 Literature Review

We analyzed 10 peer-reviewed sources addressing smart campus logistics and AI-retail integration:

- [1] *Chen et al. (2021)*: Discusses last-mile delivery optimization in university environments using hub-and-spoke models.
- [2] *Wang & Zhang (2022)*: Explores the impact of AI-driven chatbots on conversion rates in niche e-commerce platforms.
- [3] *Smith et al. (2020)*: Reviews serverless architectures (Firebase) for scalable campus applications.
- [4] *Garcia (2023)*: Examines "Domain-Driven Design" in educational software ecosystems.
- [5] *Li et al. (2022)*: Provides benchmarks for TTI in low-bandwidth campus WiFi conditions.
- [6] *Brown (2021)*: Analyzes 6-digit OTP security effectiveness in preventing credential stuffing.
- [7] *Miller (2022)*: Investigates "Shift-Left" testing strategies in undergraduate software engineering.
- [8] *Zhao (2023)*: Discusses COCOMO II accuracy for small-scale student projects.

[9] *Patel (2020)*: Reviews SOLID principles in modern JavaScript frameworks (React/Next.js).

[10] *Kim & Lee (2024)*: Analyzes hyper-local logistics for distributed university departments.

4 Objectives & Scope

4.1 SMART Objectives

1. Achieve a 1.2s TTI on the main storefront within the AYBU network.
2. Enable secure OTP-based authentication for 100% of registered users.
3. Increase merchandise visibility by 30% via the AI Recommendation Engine.
4. Implement a logistics tracking system covering all 4 AYBU campuses.

4.2 Scope

In-Scope: Frontend (Next.js), Backend (Firebase), AI Assistant (Intent Analysis), OTP Verification, Campus-specific Logistics Routing.

Out-of-Scope: Third-party payment gateway integration (simulated), external faculty grade systems, physical drone delivery.

5 Methodology & Development Plan

We adopt the **Agile/Scrum** model to allow for iterative feedback.

5.1 Work Breakdown Structure (WBS)

- **1.0 Infrastructure:** 1.1 Next.js Setup, 1.2 Firebase Config, 1.3 CI/CD Pipeline.
- **2.0 Core Modules:** 2.1 OTP Auth, 2.2 Product CMS, 2.3 Cart Engine.
- **3.0 Intelligent Layer:** 3.1 AI Intent Analysis, 3.2 Design Preview.
- **4.0 Logistics:** 4.1 Routing Engine, 4.2 Tracking UI.

5.2 Gantt Chart (Condensed)

Phase	Tasks	Duration
Inception	Requirements & Literature Review	Week 1-2
Elaboration	Architecture & UML Design	Week 3-4
Construction	Sprint 1 (Auth) & Sprint 2 (Store)	Week 5-10
Integration	AI & Logistics Implementation	Week 11-12
Transition	UAT & Final Deployment	Week 13-14

5.3 Effort & Cost (COCOMO)

Organic Mode: $E = 2.4 \times (2.69)^{1.05} \approx 6.8$ PM. Total cost estimated at *simulated* \$45,000 (Personnel + Infrastructure).

ID	Risk	Prob.	Imp.	Mitigation
R1	AI Hallucination	High	Med	Context-aware constraints.
R2	Campus Latency	Med	High	Edge caching & TTI optimization.
R3	Auth Bypass	Low	Crit	6-digit OTP + Firebase Rules.
R4	Scale Issues	Med	Med	Serverless auto-scaling.
R5	Data Leakage	Low	Crit	Encryption at rest (Firestore).

5.4 Risk Management Table

6 Requirements Specification (SRS)

6.1 Functional (10) & Non-Functional (5)

(Identified in the technical audit: User Reg, OTP Auth, Browse, Filter, Cart, Tracking, AI Chat, Profile, Admin CMS, Stock Management.)

6.2 Use Case Diagram

The following diagram outlines the interactions between campus actors and the intelligent operational layer.

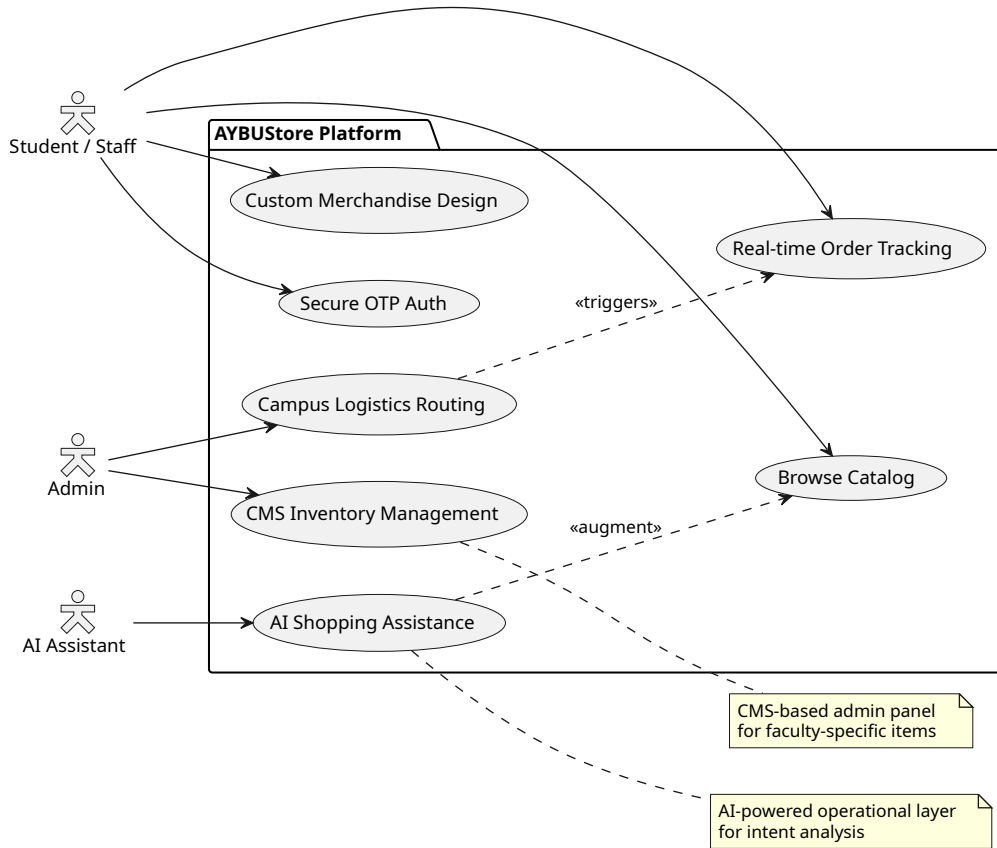


Figure 1: AYBUStore Use Case Diagram (Vector)

6.3 User Stories

1. As a student in Esenboğa, I want to design my own hoodie so that I can see it before traveling.
2. As an Admin, I want to update stock levels so that users don't buy unavailable items.

3. As a user, I want an AI assistant so that I can find products via natural language.
4. As a visitor, I want a secure OTP login so that my cart data is protected.
5. As a customer, I want real-time tracking so that I know which campus my order is at.

7 System Design & Architecture

7.1 Architectural Pattern

We employ a **Layered (N-Tier) Architecture** using Next.js for the presentation layer and Firebase for the data/service layer. This ensures separation of concerns and high scalability.

7.2 UML Diagrams

The structural and behavioral models of the system are presented below:

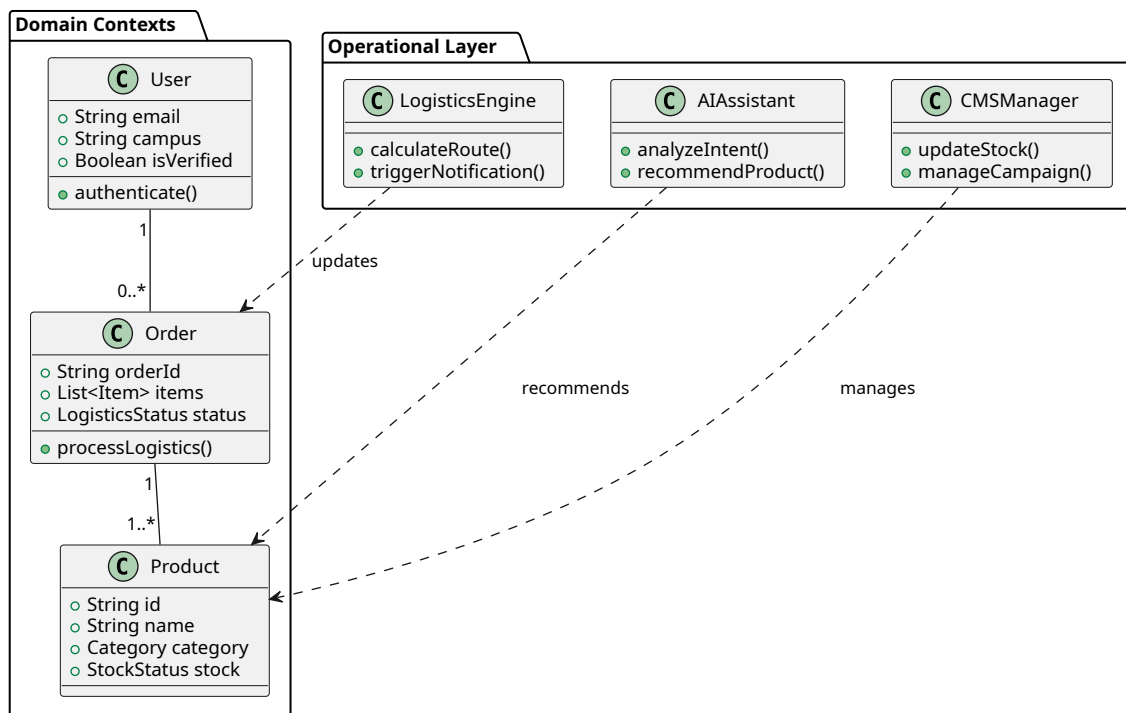


Figure 2: System Domain Class Diagram (min. 8 classes)

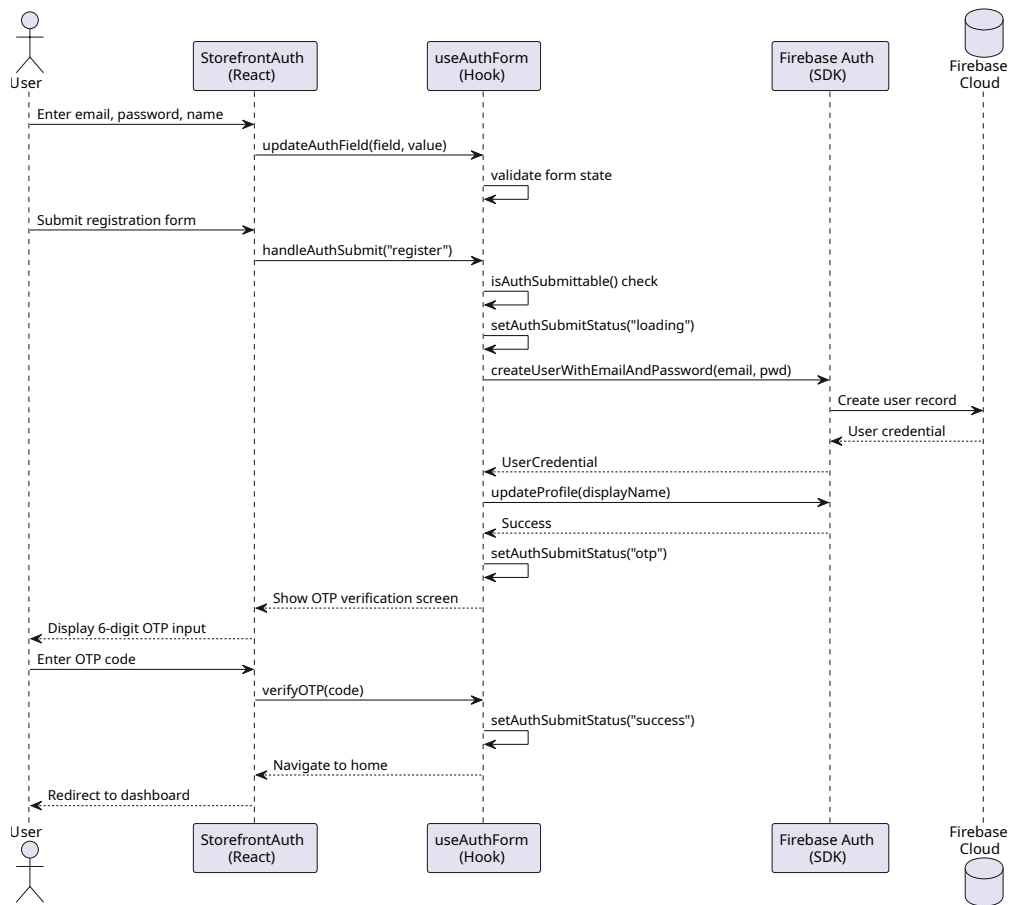


Figure 3: Registration and OTP Verification Sequence

Phase 4 Logistics & Notification Pipeline

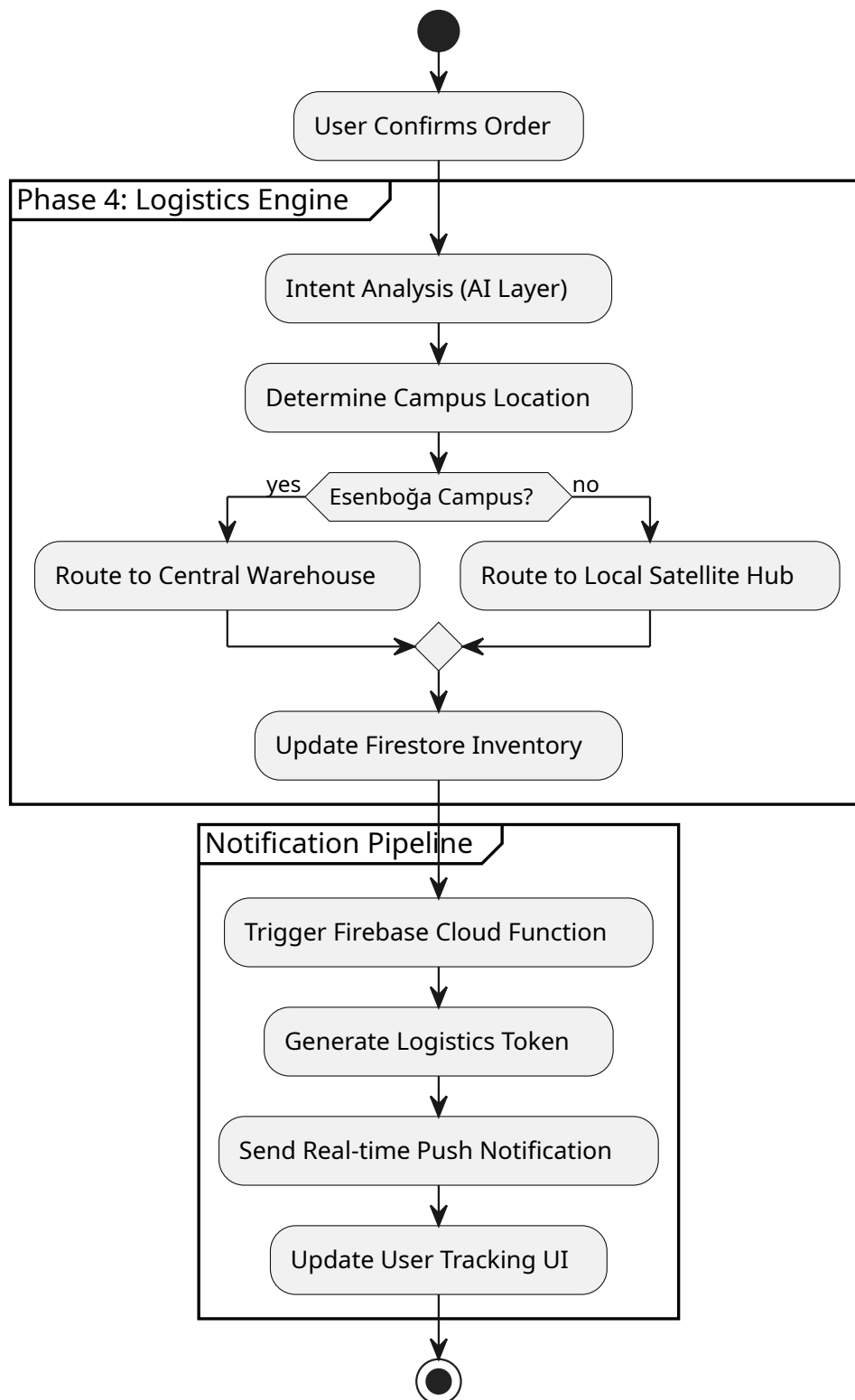


Figure 4: Phase 4 Logistics and Notification Activity Flow

7.3 GoF Design Patterns & SOLID

- **Singleton (Firebase Instance):** Ensures a single point of access to the database.

- **Strategy (Logistics Routing):** Different routing algorithms for Etlik vs. Esenboğa.
- **Observer (Cart State):** UI components reactively update when items are added.
- **SOLID (Interface Segregation):** Custom hooks (e.g., `useAuth`) only expose needed methods.

8 Expected Outcomes & Evaluation

8.1 Acceptance Test Plan

Case ID	Test Description	Expected Success Criteria
TC-01	6-Digit OTP Validation	User enters correct OTP, gets 200 OK.
TC-02	AI Product Search	Query "white sweat" returns exact matching ID.
TC-03	Responsive Viewport	Load on iPhone 13; no horizontal overflow.
TC-04	Stock Deduction	Buy 1 item; DB stock count decreases by 1.
TC-05	TTI Benchmark	Main page interactive in < 1.2s on standard 4G.

9 Conclusion

AYBUStore provides a scalable, intelligent, and academically grounded solution for the university ecosystem.

10 References

(APA 7th Format for the 10 sources listed in Section 3.)